



SECI1113: COMPUTATIONAL MATHEMATICS
20232024 – SEMESTER 2

ASSIGNMENT #3

(Deadline of Submission: **7th June. 2024; 5 pm**)

QUESTION 1

(10 MARKS)

- a) If a speedometer's absolute error is 2 mph and it measures a speed of 60 mph, what is the relative error of this measurement expressed as a percentage? (3 marks)
- b) Find the absolute error for $f'(2)$ for $f(x) = x$, given that

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \text{ and } h = 0.1.$$

(3 marks)

- c) Find the most accurate value for the following operation where each number has been rounded to certain decimal point:

$$6.232 + 7.0321 + 5.43 - 3.385$$

(4 marks)

QUESTION 2

(5 MARKS)

Given a function, $f(x) = x^3 - 6x + 1$. Find the exact value of $f(1.22)$. Then, find the absolute error and relative error when the value of $f(1.22)$ is approximated in 3 decimal points.

QUESTION 3

(18 MARKS)

Let $f(x) = 3x - e^{-x}$.

- a) Check whether the root exist in the interval:

i) $[0.1, 1.5]$

ii) $[1, 2]$

- b) Based on interval found in (a), find the root using Bisection method, approximate in 3 decimal points. Use $\varepsilon = 0.0001$

QUESTION 4

(12 MARKS)

For a non-linear function, $f(x) = \cos(x) + 2\sin(x) + x^2$, find the root using the following method:

- a) Secant method, $x_0 = 0$; $x_1 = -0.1$
- b) Newton method, $x_0 = -0.1$

Then, give some comments by comparing your results between these two methods. Do calculation in 4 decimal points and approximate the root in 2 decimal points.

Note:

1. For questions 3 and 4, you need to solve them using manual calculations and develop the Python code. Both solutions must be submitted.
2. For the Python code, write it in one file with a proper title.
3. The solution using Python contributes 15 marks.